Vargas, C, Whelan, J, Brimblecombe, J, Brock, J, Christian, M, Allender, S. Co-creation of healthier food retail environments: a systematic review to explore the type of stakeholders, their motivations and stage of engagement.

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Table S1: General description of included studies

Author, year of publication, and country	Study design, main aim, program/project name and duration of the study	Setting description and other characteristics	Participant food stores: type and ownership	MMAT Overall Score (%)
Schurman (1983)[1] Canada	• Case study • **Aim:** to improve nutritional status in remote native communities in northern Canada. • HBC Nutrition Upgrading Program	• **Population:** Inuit • **Geography/Area:** Remote and very remote	○ 64 communities ○ Community store	40
Light, et al., (1989)[2] United States	• Case-control study. • **Aim:** to test the feasibility of supermarkets as a site for consumer nutrition education. • Eat for Health • 12-month development and baseline data collection, 24-month interventions, and 12-month data analysis	• **Geography/Area:** Washington and Baltimore	○ 40 Supermarkets (20 intervention and 20 comparative) ○ Independent	40
Närhinen, et al., (1999)[3] Finland	• Case study. • **Aim:** to encourage the supermarket to take health aspects, specifically related to salt and saturated fat into consideration in their marketing • Healthier choice • 8-week intervention	• **Geography/Area:** Mikkeli	○ 1 Supermarket	40
Gittelsohn, et al., (2010)[4] Canada*	• Case-control study • **Aim:** to reduce risk factors for chronic disease in Inuit communities in the Canadian Arctic. • The Healthy Foods North (HFN) program • 6-month formative research, 18-month intervention refinement and feedback, 14 -month implementation	• **Population:** Inuit • **Geography/Area:** remote	○ 3 communities (2 intervention and 1 comparative) ○ Food stores and cooperatives ○ Chains and cooperatives	80
Gittelsohn, et al., (2010)[5] United States**	• Case-control study • **Aim:** to increase the availability of healthy foods in stores in the two intervention communities and promote healthier food choices and food preparation methods. • Healthy Foods Hawaii intervention • 8- and 10-month intervention	• **Population:** Native Hawaiian and Pacific Islander • **Geography/Area:** two islands • **Other**: low-income level and poverty	○ 2 communities ○ 4 Food stores	80

Author, year of publication, and country	Study design, main aim, program/project name and duration of the study	Setting description and other characteristics	Participant food stores: type and ownership	MMAT Overall Score (%)
Novotny, et al., (2011)[6] United States**	• Case-control study. • **Aim:** to modify the food environment of rural underserved communities to shift food availability and consumption to healthier local foods, to ultimately prevent and reduce child obesity. • Healthy Foods Hawai'I (HFH) • The intervention was comprised of four phases, each running for 6-8 weeks.	• **Population:** Native Hawaiian and Pacific Islander • **Geography/Area:** two islands • **Other**: below poverty level	○ 4 Food stores (2 intervention and 2 comparative)	100
Kolahdooz, et al. (2014)[7] Canada*	• Quasi experimental study randomly selected • **Aim:** to reduce the risk of chronic disease by improving diet and increasing physical activity. • Healthy Foods North (HFN)	• **Population:** Inuit and Inuvialuit • **Geography/Area:** Remote **Other**: below poverty level	○ 6 communities Food store	100
Young, et al (2014)[8] United States	• Case study • **Aim:** to increase consumer knowledge of the importance of a healthy diet to encourage healthy food consumption. • 18-month intervention	• **Population:** white and African American • **Geography/area:** Urban • **Other:** unemployment and poverty	○ 3 corner stores (one did not finish) ○ Independent	60
Gudzune, et al., (2015)[9] United States	• Pre- post- non-randomised intervention • **Aim:** to pilot collaborations between two urban farms with two corner stores to increase access to fresh produce in low-income neighbourhoods. • 9-week intervention	• **Geography/area:** urban • **Other:** low-income level	○ 2 corner stores	60
Ortega, et al., (2015) United States[10]	• Case-control study. • **Aim:** to improve the food environment by transforming corner stores so that they provide healthy food options to community residents. • Proyecto Mercado FRESCO • 2-year study	• **Population:** Latino • **Geography/Area:** Urban • **Other:** Food swamp	○ 8 corner stores (4 intervention and 4 comparative) ○ Independent	60
Pothukuchi (2016)[11] United States	• Case study • **Aim:** to determine if interventions could be developed to sustainably increase the availability and sales of fresh produce in corner stores in impoverished Detroit neighbourhoods. • Detroit FRESH • 6-year study	• **Geography/Area:** Detroit • **Other:** Impoverish neighbours with outdated infrastructure	○ 15 corner and 9 convenience stores ○ Independent and chain	60

Author, year of publication, and country	Study design, main aim, program/project name and duration of the study	Setting description and other characteristics	Participant food stores: type and ownership	MMAT Overall Score (%)
Winkler, et al (2016)[12] Denmark	• Case-control study. • **Aim:** to examine consumer attitudes regarding roles and responsibilities of supermarkets in health promotion and to evaluate sales effects of a healthy checkout supermarket intervention. • Project SoL • 4-week intervention	• **Geography/Area:** Bornholm Island	○ 7 supermarkets ○ Chain (3 different groups)	80
Adjoian, et al. (2017)[13] United States	• Case study. • **Aim:** to determine whether implementing the healthy checkout could increase healthy purchases without decreasing total revenue generated from this area. • Shop healthy program • 1-month intervention	• **Geography/Area:** Urban • **Other:** low income, education attainment, and health outcomes	○ 3 Supermarkets ○ Independent	80
Brimblecombe, et al., (2017)[14] Australia	• Longitudinal multi-site case study • **Aim:** to develop and test the feasibility of a monitoring and evaluation learning approach to improve the capacity of Aboriginal and Torres Strait Islander communities, and their various service providers to influence the food system to increase availability, affordability, and accessibility of healthy food. • The Good Food Systems project • 5-year project	• **Population:** First Nations • **Geography/Area:** Remote	○ 4 communities ○ Food stores	80
Thorndike, et al (2017)[15] United States	• Randomised control trial • **Aim:** to increase the visibility and quality of fresh produce in corner stores. • 5-month intervention	• **Population:** Hispanic/Latin • **Geography/Area:** Urban • **Other:** poverty	○ 6 corner stores	100
Jernigan, et al., (2018)[16] United States***	• Randomised control trial • **Aim:** to increase vegetable and fruit intake among Native Americans living within the Chickasaw and Choctaw Nation of Oklahoma. • Tribal Health and Resilience in Vulnerable Environments (THRIVE) • 5-year study	• **Population:** Native Americans • **Geography/Area:** Regional	○ 8 convenience stores ○ Community	80

Author, year of publication, and country	Study design, main aim, program/project name and duration of the study	Setting description and other characteristics	Participant food stores: type and ownership	MMAT Overall Score (%)
Young, et al., (2018)[17] United States	• Case study • **Aim:** to help corner stores sell high-quality produce by increasing supply of healthy foods and funding minor store upgrades to facilitate change. • Heights Healthy Corner Store Initiative • Two-year pilot intervention and evaluation	• **Population:** Native American	○ 5 corner stores	80
Bird Jernigan, et al., (2019)[18] United States***	• Cluster-controlled trial. • **Aim:** to assess a healthy retail intervention in Tribal convenience stores in Oklahoma. • **Program/project name:** Tribal Health and Resilience in Vulnerable Environments (THRIVE) • 5-year study	• **Population:** Native Americans • **Geography/Area:** Rural	○ 8 convenience stores (4 intervention and 4 comparative) ○ Tribal owned	80
Brimblecombe, et al., (2020)[19] Australia	• Randomised control trial • **Aim:** to evaluate the impact of a strategy co-designed with industry to reduce merchandising of discretionary products on customer purchasing and business performance. • Healthy Stores 2020 • 12-week intervention	• **Population:** First Nations • **Geography/Area:** Remote	○ 20 community stores (10 intervention and 10 comparative)	100
Fehring, et al., (2019)[20] Australia	• Case-control study. • **Aim:** to create supportive environments to reduce sugary drink consumption and increase water consumption by partnering with remote Aboriginal and Torres Strait Islander communities in Cape York. • The Healthy Communities Project for health. • 2-month planning, 12-month implementation	• **Population:** Aboriginal • **Geography/Area:** Remote	○ 4 community stores ○ independent	80
Young, et al (2020)[21] New Zealand	• Case-control study • **Aim:** to place the healthier breakfast cereal products within each cereal segment more prominently on shelves, ideally at adult eye- level. • 36-week pilot intervention trial: 12-week baseline, 12-week intervention, 12-week follow-up	• **Geography/Area:** Urban • **Other:** low- to high-income	○ 6 supermarkets (3 intervention and 3 comparative) ○ Chain	100
Bogomolova, et al., (2021)[22] Australia	• Case study • **Aim:** to map out and examine the process of co-designing and implementing a programme that encourages healthier choices in a real supermarket.	• **Geography/Area:** Urban	○ 1 supermarket ○ Independent	60

Author, year of publication, and country	Study design, main aim, program/project name and duration of the study	Setting description and other characteristics	Participant food stores: type and ownership	MMAT Overall Score (%)
	To evaluate the effectiveness of a co-designed programme for improving the healthfulness of food choices in a real supermarket.			
Rollins, et al (2021)[23] United States	• Mix-methods • **Aim:** To increase access to healthy foods in a south Metropolitan Atlanta community. • The REACH-HI Healthy Corner Store Initiative	• **Population:** African American • **Geography/Area:** Urban • **Other:** low-income	○ 11 corner stores	60

Table S2: Quality appraisal of included studies

First author, year of publication	S1	S2	1.2	1.2	1.3	1.4	1.5	2.1	2.2	2.3	2.4	2.5	3.1	3.2	3.3	3.4	3.5	4.1	4.2	4.3	4.4	4.5	5.1	5.2	5.3	5.4	5.5	Overall score (%)
Schurman (1983)	Y	Y	Y	N	Y	N	?																					40
Light, et al., (1989)	Y	Y											?	Y	N	Y	?											40
Närhinen, et al., (1999)	Y	Y											Y	?	Y	N	N											40
Gittelsohn, et al., (2010)	Y	Y											Y	Y	Y	Y	?											80
Gittelsohn, et al., (2010)	Y	Y											Y	Y	Y	Y	?											80
Novotny, et al., (2011)	Y	Y											Y	Y	Y	Y	Y											100
Kolahdooz, et al. (2014)	Y	Y						Y	Y	Y	Y	Y																100
Young, et al (2014)	Y	Y											Y	Y	Y	N	?											60
Gudzune, et al., (2015)	Y	Y											N	Y	Y	Y	N											60
Ortega, et al., (2015)	Y	Y											Y	Y	Y	N	?											60
Pothukuchi 2016)	Y	Y	Y	Y	Y	N	?																					60

First author, year of publication	S1	S2	1.2	1.2	1.3	1.4	1.5	2.1	2.2	2.3	2.4	2.5	3.1	3.2	3.3	3.4	3.5	4.1	4.2	4.3	4.4	4.5	5.1	5.2	5.3	5.4	5.5	Overall score (%)
Winkler, et al (2016)	Y	Y											?	Y	Y	Y	Y											80
Adjoian, et al. (2017)	Y	Y											Y	Y	Y	?	Y											80
Brimblecombe, et al., (2017)	Y	Y																Y	Y	Y	?	Y						80
Thorndike, et al (2017)	Y	Y						Y	Y	Y	Y	Y																100
Jernigan, et al., (2018)	Y	Y						Y	Y	Y	?	Y																80
Young, et al., (2018)	Y	Y											Y	?	Y	Y	Y											80
Bird Jernigan, et al., (2019)	Y	Y											Y	Y	Y	Y	?											80
Fehring, et al., (2019)	Y	Y											?	Y	Y	Y	Y											80
Brimblecombe, et al., (2020)	Y	Y						Y	Y	Y	Y	Y																100
Young, et al (2020)	Y	Y											Y	Y	Y	Y	Y											100
Bogomolova, et al., (2021)	Y	Y											Y	Y	Y	?	?											60
Rollins, et al 2021)	Y	Y	Y	Y	Y	N	N						N	Y	Y	Y	N						Y	Y	Y	?	N	60

Table S3: Mixed methods appraisal tool (MMAT) criteria

Category of study designs	Methodological quality criteria
Screening questions (for all types)	S1. Is there a clear qualitative and/or quantitative research question (or research objective)?
	S2. Do the collected data allow answering (meeting) the research question (objective)? E.g., consider whether the follow-up period was long enough for the outcome to occur (concerning longitudinal studies or study components).
1. Qualitative	1.1. Is the qualitative approach appropriate to answer the research question?
	1.2. Are the qualitative data collection methods adequate to address the research question?
	1.3. Are the findings adequately derived from the data?
	1.4. Is the interpretation of results sufficiently substantiated by data?
	1.5. Is there coherence between qualitative data sources, collection, analysis and interpretation?
2. Quantitative randomized controlled (trials)	2.1. Is randomization appropriately performed?
	2.2. Are the groups comparable at baseline?
	2.3. Are there complete outcome data?
	2.4. Are outcome assessors blinded to the intervention provided?
	2.5 Did the participants adhere to the assigned intervention?
3. Quantitative non-randomized	3.1. Are the participants representative of the target population?
	3.2. Are measurements appropriate regarding both the outcome and intervention (or exposure)?
	3.3. Are there complete outcome data?
	3.4. Are the confounders accounted for in the design and analysis?
	3.5. During the study period, is the intervention administered (or exposure occurred) as intended?
4. Quantitative descriptive studies	4.1. Is the sampling strategy relevant to address the research question?
	4.2. Is the sample representative of the target population?
	4.3. Are the measurements appropriate?
	4.4. Is the risk of nonresponse bias low?
	4.5. Is the statistical analysis appropriate to answer the research question?
5. Mixed methods	5.1. Is there an adequate rationale for using a mixed methods design to address the research question?
	5.2. Are the different components of the study effectively integrated to answer the research question?
	5.3. Are the outputs of the integration of qualitative and quantitative components adequately interpreted?
	5.4. Are divergences and inconsistencies between quantitative and qualitative results adequately addressed?
	5.5. Do the different components of the study adhere to the quality criteria of each tradition of the methods involved?

Reference: Hong QN, Pluye P, Fàbregues S, Bartlett G, Boardman F, Cargo M*, et al.* Mixed Methods Appraisal Tool: User guide 2018. Retrieved on [21 January 2021] from: http://mixedmethodsappraisaltoolpublic.pbworks.com/w/file/fetch/127916259/MMAT_2018_criteria-manual_2018-08-01_ENG.pdf.

Table S4: Author's Reflections and recommendations

	Themes and subthemes	Examples extracted from text
Reflections	○ Enhancement: ○ implementation[1, 2, 6, 9, 10, 17, 20, 22] ○ design [4, 14, 20]	○ A major element of the *Eat for Health* study was its collaborative nature, with joint planning, development, and management.[2] ○ The development of HFN purposefully emphasized stakeholder involvement through formative research, small group presentations, community workshops, and training as a means of understanding the needs of the target population, addressing them in a culturally relevant manner, encouraging participation, and building consensus among stakeholders.[4]
	○ Empowerment: ○ community[4, 10, 19, 22, 23] ○ retailers[3, 22]	○ Three main facilitators for a successful store conversion. 1) strong community involvement through all phases of the intervention; 2) Include a partner that understands the business; 3) capacity building and social marketing as patrons' drivers for collaboration.[10] ○ Staff's perspective needs to be actively considered as staff offer much-needed expertise in the feasibility aspect of any new programme ideas.[22] During the intervention it was important give autonomy to the supermarket's management to apply new ideas, instead of strictly keeping to the original plan.[3]
	○ Impacts on project sustainability[6, 14, 21]	○ Working with multiple stakeholders is challenging, however, the integration of community-based organizations into intervention delivery enhanced implementation and likelihood of sustainability.[6]

	Themes and subthemes	Examples extracted from text
	○ Strength of relationships: ○ community[17] ○ between sectors [18, 19, 21] ○ growing partnerships[3]	○ Un expected outcomes included improved connections to community members and customer's increased knowledge of fresh produce.[17] ○ This study used co-design to enhance the likely fiscal sustainability of the intervention. This process allowed a strong working relationship to be built with the retail partner, which facilitated intervention delivery, access to sales and promotional data, and possible future research opportunities.[21] ○ The project promoted new partnerships (voluntary heart health organisation and the supermarket). The municipal food control authority served as a bridge between them. This study was thought useful to test the possibility of co-operation between food control and supermarkets.[3]
Recommendations	○ Prolonged time for the intervention[5, 14, 22]	○ A more intense program of longer duration, supported by policy changes, would likely have broader impacts.[5]
	○ Extended stakeholders' diversity[5, 7, 14, 17]	○ There is greater possibility for sustainability if the programs partner with community-based institutions such as schools and stores.[7]
	○ Greater capacity building[10, 13, 17]	○ This type of intervention might be unsustainable in a long term as checkout lines are often stocked by snack vendors or ongoing staff training on healthy products might be needed.[13]
	○ Specific conditions of the setting[5, 6, 8, 9, 16]	○ Setting infrastructure and local promotion are two influential factors for success.[9]
	○ Policy support[5, 15]	○ In the future, the US Department of Agriculture might consider requiring WIC- and SNAP-certified stores to display fruits/vegetables at the front of the stores

	Themes and subthemes	Examples extracted from text
		and to provide education for store owners about stocking and maintaining fresh produce.[15]
	○ More intensive programs[10, 22]	○ Future programmes should cover more SKUs and product categories to guide more shoppers more often. Running programme elements in several waves to refresh and revive shoppers' experiences, while ensuring strong continuity in the whole programme and fidelity to original objectives.[22]
	○ Consideration of business needs[17]	○ Addressing structural factors within business practices is critical to project success.[17]

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